STARTUP & ENGINEERING MASTER BLUEPRINT

AI Healthcare Triage, Infrastructure Discovery & Cost Transparency Platform

End-to-End Technical Architecture, API Stack, Data Collection Pipeline, Database Schema & Execution Roadmap (Hinglish & Technical Edition)

1. Executive Summary & Core Value Proposition

The Problem (Asli Dikkhat): India mein jab kisi patient ko critical bimari (jaise Cancer/Oncology, Heart Attack/STEMI, Stroke, ya Renal Failure) hoti hai, toh family ko yeh nahi pata hota ki kis hospital mein kaun si **medical machine (Equipment)**, **super-specialist**, aur **treatment capability** chalu halat mein hai. Log ek hospital se dusre hospital bhatakte hain, jisse *Golden Hour* aur paisa dono barbaad hote hain.

The Solution (Tumhara Platform): Ek AI-powered platform jahan user ghar baithe patient ke symptoms bataye ya diagnostic report (Biopsy, CT, Blood Test) upload kare. System AI ke zariye report padh kar batayega ki **kaun si machine/department chahiye**, user ke GPS location ke aas-paas **kis hospital mein woh exact equipment available hai**, aur wahan **Government Scheme vs. Private Package mein kitna kharcha aayega**.

> **Unique Selling Proposition (USP):** Practo aur Lybrate *Doctor-centric* hain (kaun sa doctor kahan baithta hai). Tumhara platform **Infrastructure & Capability-centric** hai—jo patient ki condition ko seedha hospital ke *Equipment Inventory (PET-CT, LINAC, Cath Lab, ECMO, ICU)* aur *Real-Time Cost/Insurance* se match karta hai.

2. End-to-End System Workflow (Kaise Kaam Karega?)

1. **Multimodal User Input:** User app/website par voice (Hindi/English/Regional) mein problem bolta hai ya prescription/report ki photo upload karta hai.
2. **OCR & Medical Entity Extraction:** Vision AI report se key medical entities extract karta hai (e.g., *"Stage IIA Non-Small Cell Lung Cancer"* ya *"Acute Anterior Wall MI"*).
3. **AI Triage & Infrastructure Mapping (RAG Engine):** AI Medical Knowledge Graph ke through decide karta hai ki is bimari ke liye kya mandatory hai (e.g., *"Needs Tertiary Oncology Center with Bronchoscopy + PET-CT + Linear Accelerator (LINAC) + Surgical Oncologist"*).
4. **Geospatial Filtering & Routing:** Database user ke current location se radius search karta hai aur un hospitals ko hata deta hai jahan woh machine nahi hai (bhale hi woh 2 km paas kyun na ho), aur sabse nazdeeki *capable* hospital dikhata hai.
5. **Cost & Scheme Transparency Engine:** Shortlisted hospitals ke samne 3 price brackets dikhte hain: **Govt/Trust Rate**, **PM-JAY / State Scheme Coverage**, aur **Private Corporate Estimate**.

3. Complete Technology Stack (Full Tech Stack)

Is project ko scalable startup aur high-grade engineering product banane ke liye modern microservices-ready stack use karna hoga:

Architecture Layer	Recommended Technology	Kyun Use Karein? (Purpose in Project)
Frontend (Web & Mobile)	Next.js 14 (React), Tailwind CSS, React Native (Expo)	Fast SEO-friendly web app + Android/iOS app ek hi codebase ke logic se; multilingual UI support.
Backend Core API	FastAPI (Python) + Node.js (Express)	FastAPI AI/ML models, OCR, aur RAG pipeline ko lightning-fast handle karta hai; asynchronous API support.
Primary Database	PostgreSQL + **PostGIS Extension**	Hospitals, equipments, aur packages ka relational data + PostGIS se exact GPS radius aur nearest hospital query.
Search & Filtering	Typesense ya Elasticsearch	Typo-tolerant instant search (agar user "kanser" ya "hart atack" likhe toh bhi sahi specialty match ho).
Vector DB (For AI)	Qdrant ya Pinecone / pgvector	Standard Treatment Guidelines (STGs) aur Disease-to-Equipment mapping store karne ke liye.
Caching & Queues	Redis + Celery	Frequent searches ko cache karne aur background mein hospital data scraping jobs chalane ke liye.
Cloud & Hosting	AWS (EC2, S3, RDS) ya Vercel + Supabase	S3 mein encrypted medical reports store hongi; Supabase initial MVP ke liye free PostgreSQL + PostGIS deta hai.

4. Complete API Directory (Kaun-Kaun Se APIs Lagenge?)

Is project ko banane ke liye tumhe har cheez zero se code nahi karni hai. Niche diye gaye APIs ko integrate karke tumhara system enterprise-level ban jayega:

Category	API / Service Name	Exact Use Case & Integration Role
Medical Report OCR & Vision AI	Google Gemini 2.5 Flash / Pro API (Multimodal) ya GPT-4o Vision	Handwritten doctor prescription, biopsy PDF, aur blood reports ki photo padh kar structured JSON nikalne ke liye.
Indian Voice & Language API	**Bhashini API** (Govt of India) ya Sarvam AI / OpenAI Whisper	Gaon ya chhote shehar ka user Hindi, Bhojpuri, Bengali, ya Tamil mein apni bimari bole, toh usko text mein badalne ke liye.

Category	API / Service Name	Exact Use Case & Integration Role
Maps, Traffic & Routing API	Google Maps Routes API / Mappls (MapmyIndia) / OpenRouteService	Sirf air-distance nahi, balki **real-time ambulance travel time (ETA)** nikalne ke liye. Mappls India ke gali-mohalle ke liye best hai.
Healthcare Public Registry API	**ABDM (Ayushman Bharat Digital Mission) Sandbox APIs**	Health Facility Registry (HFR) aur Healthcare Professionals Registry (HPR) se verified hospitals aur doctors ka base data lene ke liye.
Medical Terminology API	WHO ICD-11 API & SNOMED-CT	Bimariyon ke aam naam ko standard medical codes (ICD codes) mein badalne ke liye taaki insurance packages match ho sakein.
Communication & Alerts	WhatsApp Business Cloud API / Twilio	User ko WhatsApp par seedha shortlisted hospitals ki list, estimated kharcha, aur navigation link bhejne ke liye.

5. All-India Data Collection Strategy (Sabse Bada Secret)

Sabse bada sawal: **"India ke saare hospitals, unke equipments, aur unke rates ka data kahan se aayega?"** Isko 4-Layer Data Pipeline se solve kiya jata hai:

Layer 1: Government Open Datasets (Zero Cost Base Data)

- **ABDM Health Facility Registry (HFR):** Government of India ke ABDM portal par hazaron registered hospitals ka official naam, address, pin code, ownership type (Govt/Private), aur specialties available hain.
- **PM-JAY (Ayushman Bharat) Empanelled Hospital List:** Public portal (`hospitals.pmjay.gov.in`) par har state aur district ke hisab se empanelled hospitals aur unke active clinical specialties (jaise Medical Oncology, Radiation Oncology, Neurosurgery) ki public list milti hai.
- **NABH & ROHINI Registry:** NABH accredited hospitals aur IRDAI ke ROHINI registry se insurance-ready hospitals ka verified data milta hai.
- **National Health Profile & Data.gov.in:** State-wise hospital directories aur district hospital infrastructure datasets.

Layer 2: Standard Cost & Package Baseline Data (Price Estimation Ke Liye)

Private hospitals apna exact bill pehle nahi batate, lekin tum **3-Tier Cost Estimation Algorithm** se 85–90% accurate estimate de sakte ho:

- **Tier 1 (Govt / Scheme Baseline): PM-JAY Health Benefit Package (HBP 2.2)** aur **CGHS (Central Govt Health Scheme)** ke official rate charts public PDFs mein available hain (jisme 1,900+ surgeries aur chemo/radiation cycles ke exact rates likhe hain).
- **Tier 2 (Trust / Semi-Private Multiplier):** CGHS base rate × `1.5x to 2.2x` (Charitable aur mid-sized NABH hospitals ka standard estimation).
- **Tier 3 (Corporate Hospital Multiplier):** CGHS base rate × `3.0x to 5.0x` + Corporate hospitals ke public tariff PDFs (jaise Tata Memorial, Apollo, Max, Fortis ke website disclosures aur insurance TPA rate cards).

Layer 3: Automated Web Scraping & LLM Extraction Pipeline

- **Tools:** Python (Playwright, BeautifulSoup, Scrapy) + Firecrawl API.
- **Kaise Karein:** Bade aur mid-size hospitals ki websites ke *"Facilities"*, *"Departments"*, aur *"Technology"* pages ko scrape karo. Raw text ko LLM (Gemini Flash) mein bhej kar prompt do: *"Extract a JSON array of all diagnostic and surgical equipment mentioned on this page (e.g., 3T MRI, PET-CT, Da Vinci Robot, LINAC, ECMO, Cath Lab)."*

Layer 4: Hospital Onboarding & Crowdsourced Verification (B2B Portal)

- Har hospital ko ek **"Claim & Verify Hospital Dashboard"** do. Jab kisi hospital mein machine kharab ho (downtime) ya nayi machine aaye, toh hospital ka PRO/Admin ek toggle button se status `Active / Under Maintenance` kar sake.
- Patients jab kisi hospital mein jayein, toh unse 1-click feedback lo: *"Kya aapko yahan PET-CT machine chalu mili? Aur kitna kharcha aaya?"* (Waze/Google Maps jaisa crowdsourced verification).

6. Core Database Schema Design (PostgreSQL + PostGIS)

Tumhara database normal directory jaisa nahi hona chahiye. Niche diya gaya relational schema is project ki reedh ki haddi (backbone) hai:

```
-- 1. HOSPITALS TABLE (Geospatial Enabled) CREATE TABLE hospitals ( hospital_id
UUID PRIMARY KEY, name VARCHAR(255) NOT NULL, hospital_type VARCHAR(50), --
'Government', 'Trust', 'Corporate', 'Nursing_Home' tier_category VARCHAR(20), --
'Primary', 'Secondary', 'Tertiary_SuperSpecialty' address TEXT, city VARCHAR(100),
state VARCHAR(100), pincode VARCHAR(10), location GEOGRAPHY(Point, 4326), --
PostGIS GPS Coordinates (Lat, Long) abdm_hfr_id VARCHAR(100), nabh_accredited
BOOLEAN DEFAULT FALSE, pmjay_empanelled BOOLEAN DEFAULT FALSE, emergency_24x7
BOOLEAN DEFAULT TRUE, contact_phone VARCHAR(20), last_verified_at TIMESTAMP ); --
2. EQUIPMENT & INFRASTRUCTURE INVENTORY (Core USP) CREATE TABLE hospital_equipment
( id UUID PRIMARY KEY, hospital_id UUID REFERENCES hospitals(hospital_id),
equipment_code VARCHAR(50), -- e.g., 'PET_CT', 'LINAC_RADIATION', 'CATH_LAB',
'ECMO', 'MRI_3T' equipment_name VARCHAR(150), operational_status VARCHAR(30), --
'OPERATIONAL', 'MAINTENANCE', 'UNAVAILABLE' total_units INT DEFAULT 1,
avg_wait_time_hours INT, updated_at TIMESTAMP ); -- 3. DISEASE TO INFRASTRUCTURE
KNOWLEDGE GRAPH (AI Mapping Table) CREATE TABLE condition_requirements (
condition_icd_code VARCHAR(20) PRIMARY KEY, -- e.g., 'C34' for Lung Cancer
condition_name VARCHAR(255), specialty_required VARCHAR(100), -- e.g., 'Medical &
Radiation Oncology' mandatory_equipment JSONB, -- ["PET_CT", "BIOPSY_UNIT",
"LINAC_RADIATION"] optional_equipment JSONB, -- ["ROBOTIC_SURGERY",
"ICU_VENTILATOR"] urgency_level VARCHAR(30) -- 'GOLDEN_HOUR', 'URGENT_48H',
'PLANNED' ); -- 4. TREATMENT PACKAGES & COST TRANSPARENCY CREATE TABLE
treatment_costs ( package_id UUID PRIMARY KEY, hospital_id UUID REFERENCES
hospitals(hospital_id), condition_icd_code VARCHAR(20) REFERENCES
condition_requirements(condition_icd_code), procedure_name VARCHAR(255), -- e.g.,
'Radical Mastectomy + Chemo Cycle' min_estimated_cost INR NUMERIC(10,2),
max_estimated_cost INR NUMERIC(10,2), pmjay_package_rate NUMERIC(10,2),
cashless_insurers JSONB -- ["Star Health", "HDFC Ergo", "Ayushman Bharat",
"Swasthya Sathi"] );
```

7. AI Triage & Matching Engine Architecture (How AI Works Here)

Is project mein AI sirf chatbot nahi hai; AI ek **Clinical Routing Decision Engine** hai. Isko गलत सलाह (hallucination) se bachane ke liye **Hybrid RAG + Rule Engine** use karna hoga:

1. **Step 1 — Document Ingestion & Privacy:** Jab user medical report upload karta hai, Python script pehle Patient ka Naam aur Phone Number mask (hide) karti hai (DPDP Act privacy ke liye).
2. **Step 2 — Structured JSON Extraction:** Multimodal LLM ko strict system prompt diya jata hai:

```
{ "detected_condition": "Suspected Carcinoma Cervix Stage IIB", "icd_11_code":
"2C77", "urgency": "HIGH_PRIORITY", "required_diagnostics": ["MRI Pelvis", "PET-
CT Whole Body", "Histopathology Biopsy"], "required_treatment_infra": ["Linear
Accelerator (LINAC)", "Brachytherapy Unit", "Daycare Chemotherapy"],
"required_specialists": ["Gynecologic Oncologist", "Radiation Oncologist"] }
```

3. **Step 3 — PostGIS + Equipment Match Scoring:** Backend SQL query user ke 50 km (ya state-level) radius mein hospitals nikal kar har hospital ko **Capability Match Score (0 to 100%)** deti hai:
   - **50% Weightage:** Kya hospital mein saari mandatory machines (e.g., LINAC + Brachytherapy) `OPERATIONAL` hain?
   - **25% Weightage:** Kya user ke budget ya Government Card (Ayushman Bharat) mein woh hospital aata hai?
   - **25% Weightage:** Travel time (ETA) aur NABH accreditation.

8. Step-by-Step Implementation Roadmap (0 to Launch Plan)

Poore India ka data ek din mein ikattha karne ki koshish mat karo. Isko 4 phases mein build karo:

Phase & Timeline	Milestone Focus	Actionable Deliverables (Kya Banana Hai)
Phase 1: Weeks 1–3 (Data & Core DB)	**1 Metro City + 3 Critical Specialties**	Apne shehar (e.g., Kolkata, Delhi-NCR, ya Mumbai) ke top 100 hospitals (Govt + Trust + Corporate) ka data PostgreSQL + PostGIS mein dalo. Sirf 3 high-pain areas chuno: **Oncology (Cancer), Cardiology (Heart), aur Nephrology (Dialysis)**.
Phase 2: Weeks 4–6 (AI & Backend)	**AI Report Reader & Matching API**	FastAPI backend banao. Gemini Vision API se prescription/biopsy report padhne ka pipeline jodo aur Disease-to-Equipment mapping table se connect karo.
Phase 3: Weeks 7–9 (Frontend & UX)	**Web App + WhatsApp Bot MVP**	Next.js par clean UI banao jahan user report upload kare, map par Equipped Hospitals dekhe, aur cost compare kare. Sath mein WhatsApp bot jodo taaki non-tech-savvy log bhi use kar sakein.
Phase 4: Weeks 10–12 (Pilot & Scale)	**Hospital Onboarding & Live Testing**	Local diagnostic centers aur 5–10 mid-sized hospitals se baat karke unka exact package rate dalo. Startup India aur ABDM Sandbox ke liye apply karo.

9. Legal, Compliance & Safety Guidelines (Bahut Zaroori)

Healthcare startup mein legal safety sabse pehle aati hai. Apne platform mein yeh 3 rules hamesha rakhna:

- **No Medical Prescription (Triage Only):** Tumhara AI kabhi bhi dawai (medicine) prescribe nahi karega. App par clear disclaimer hoga: *"This platform provides hospital infrastructure discovery and cost estimation based on clinical reports. It does not replace a doctor's medical diagnosis."*
- **DPDP Act 2023 Compliance (Data Privacy):** Patient ki medical reports sensitive personal data hain. Upload karne se pehle user se explicit consent checkbox lo aur files ko encrypted cloud storage mein rakho.

- **Estimated Price Ranges:** Kabhi bhi fixed price (jaise ₹1,42,000) mat dikhao; hamesha range dikhao (e.g., *Estimated ₹1.2L – ₹1.8L depending on clinical complications and room category*).

10. Monetization & Startup Revenue Model (Paisa Kaise Banega?)

- **B2B Patient Facilitation Fee (Primary Revenue):** Mid-sized NABH hospitals aur corporate chains ko high-intent patients (cancer surgery, radiation, bypass surgery, knee replacement) chahiye hote hain. Jab tumhare platform se verified patient wahan admit hota hai, toh hospital **5% to 12% marketing/facilitation commission** deta hai (Patient se ₹0 liya jata hai).
- **Diagnostic Equipment Booking Commission:** PET-CT (₹12,000–₹25,000), 3T MRI, aur Biopsy tests par diagnostic centers **15% to 25% referral fee** dete hain.
- **Insurance & Medical Financing Tie-ups:** Jin patients ke paas cashless insurance nahi hai, unhe No-Cost Medical EMI startups (jaise CarePal, SaveIN, ImpactGuru) se connect karke commission milta hai.
- **SaaS Dashboard for Hospitals:** Hospitals ko premium analytics dena ki unke इलाके (50 km radius) mein kis bimari aur kis machine ki sabse zyada search ho rahi hai.

11. Deep-Dive API Implementation & Integration Payloads (Technical Reference)

Jab tum backend ya AI agent mein real APIs connect karoge, toh har API ka exact endpoint flow, input request, aur JSON response is tarah kaam karega:

11.1 Vision AI Medical Report OCR API (Gemini 2.5 Flash / GPT-4o Vision)

Role: Patient ki uploaded Biopsy, CT Scan, ECG, ya Blood Report ki photo/PDF padh kar usme se bimari (ICD-11 code) aur zaroori medical machines (Equipment Codes) nikalna.

```
// POST /api/v1/triage/analyze-report // System Instruction to Vision LLM: "You are a Clinical Infrastructure Triage Engine. Analyze the uploaded medical report after PII masking. Do NOT prescribe medication. Output strictly in the following JSON schema mapping the diagnosis to mandatory hospital equipment codes: [PET_CT, LINAC_RADIATION, BRACHYTHERAPY, CATH_LAB, ECMO, MRI_3T, DIALYSIS_CRRT, BIOPSY_HISTOPATH, NEURO_MICRO, VENTILATOR_ICU]." // Expected Structured JSON Output from API: { "patient_pii_masked": true, "detected_condition": "Non-Small Cell Lung Carcinoma (Stage IIA)", "icd_11_code": "2C25", "urgency_level": "URGENT_48H", "specialty_required": "Medical, Surgical & Radiation Oncology", "mandatory_equipment": ["PET_CT", "LINAC_RADIATION", "BIOPSY_HISTOPATH"], "optional_equipment": ["VENTILATOR_ICU"], "clinical_warning": "Requires single-roof Tertiary Cancer Center with active Linear Accelerator (LINAC) and PET-CT staging to avoid inter-hospital referral delays." }
```

11.2 Bhashini API (Govt of India ULCA) — Vernacular Voice-to-Text & Translation

Role: Gaon ya tier-2/3 city ke log jab Hindi, Bengali, Bhojpuri, ya Tamil mein bolenge (e.g., *"Mere mareez ko seene mein tej dard hai aur paseena aa raha hai"*), toh Bhashini pipeline (`ASR` + `NMT`) usko clinical English query mein badal dega.

```
// Bhashini ULCA Pipeline Payload (POST https://dhruva-
api.bhashini.gov.in/services/inference/pipeline) { "pipelineTasks": [ { "taskType":
"asr", "config": { "language": { "sourceLanguage": "hi" }, "audioFormat": "wav" }
}, { "taskType": "translation", "config": { "language": { "sourceLanguage": "hi",
"targetLanguage": "en" } } } ], "inputData": { "audio": [{ "audioContent": "
<BASE64_ENCODED_VOICE_NOTE>" }] } }
```

11.3 Mappls (MapmyIndia) / Google Routes API — Golden Hour Ambulance ETA

Role: Hawa mein distance (air-distance) dekhne se kaam nahi chalta; traffic ke hisab se exact **Ambulance Travel Time (ETA in minutes)** nikalna zaroori hai.

```
// GET /api/v1/routing/eta-matrix?
origin=22.5726,88.3639&destinations=22.5800,88.4600|22.5411,88.3426 // Returns
real-time traffic-adjusted duration_in_traffic (seconds) and distance (meters) //
PostGIS + Routing Combined Query: SELECT h.hospital_id, h.name, h.hospital_type,
ROUND((ST_Distance(h.location, ST_SetSRID(ST_MakePoint(88.3639, 22.5726),
4326)::geography) / 1000)::numeric, 2) AS distance_km FROM hospitals h WHERE
ST_DWithin(h.location, ST_SetSRID(ST_MakePoint(88.3639, 22.5726), 4326)::geography,
50000) ORDER BY distance_km ASC;
```

11.4 ABDM (Ayushman Bharat Digital Mission) Health Facility Registry (HFR) API

Role: Hospital ka official registration, NABH accreditation, aur ownership verify karne ke liye.

- **Sandbox Endpoint:** `https://facilitysbx.abdm.gov.in/v1.5/facilities/search`
- **Key Fields Extracted:** `facilityId` (HFR ID), `facilityName`, `ownershipCode` (Govt/Private/Trust), `address.pincode`, `latitude`, `longitude`, `specialties`, aur `ayushmanBharatEmpanelled`.

12. Automated Data Collection & Scraping Pipeline (Step-by-Step Code Logic)

All-India hospitals ka database banane ke liye tumhe haath se entry nahi karni hai. Niche diya gaya **Python + LLM Scraping Pipeline** kisi bhi hospital ki website ya public PDF se equipment aur package rates nikal kar seedha PostgreSQL mein daal dega:

```
# file: data_pipeline/hospital_equipment_scraper.py # Automated Pipeline: Scrape Hospital Website -> LLM Structured Extraction -> PostgreSQL Insert import json from pydantic import BaseModel from typing import List class EquipmentItem(BaseModel): equipment_code: str # e.g., PET_CT, LINAC_RADIATION, CATH_LAB, MRI_3T, ECMO equipment_name: str # e.g., Siemens Biograph PET-CT operational_status: str # Default: OPERATIONAL class HospitalInfrastructureProfile(BaseModel): hospital_name: str city: str nabh_accredited: bool pmjay_accepted: bool extracted_equipment: List[EquipmentItem] # Step 1: Crawl hospital's /facilities, /departments/oncology, /cardiology pages using Playwright/Firecrawl # Step 2: Pass cleaned markdown text to Gemini 2.5 Flash with response_schema=HospitalInfrastructureProfile # Step 3: Apply 3-Tier Cost Multiplier against Official CGHS / PM-JAY HBP 2.2 Base Tariff CSV: def calculate_3_tier_cost_brackets(cghs_base_rate: float, hospital_type: str, room_modifier: float = 1.0): if hospital_type == "Government": return {"min": 0, "max": round(cghs_base_rate * 0.35 * room_modifier, -2)} elif hospital_type == "Trust": return {"min": round(cghs_base_rate * 1.5 * room_modifier, -2), "max": round(cghs_base_rate * 2.2 * room_modifier, -2)} else: # Corporate Super-Specialty return {"min": round(cghs_base_rate * 3.0 * room_modifier, -2), "max": round(cghs_base_rate * 4.8 * room_modifier, -2)}
```

13. Complete Antigravity Master Prompt Playbook (Copy-Paste Sequence)

Apne AI Coding Agent (Antigravity) se **100% working, production-grade website (MedRadar AI / ArogyaPath AI)** banwane ke liye niche diye gaye **8 Step-by-Step Prompts** ko ek-ek karke sequence mein copy-paste karo. Ek hi prompt mein sab mat dena—pehle Step 1 build hone do, phir Step 2 dena.

STEP 1 PROMPT: Core Architecture, Clinical Design System & Real Database Engine

```
Act as a Principal Full-Stack Healthcare Architect and Elite UI/UX Designer. We are building "MedRadar AI" (ArogyaPath AI) — an end-to-end working AI Healthcare Triage, Hospital Infrastructure Discovery, and Cost Transparency Platform. CRITICAL INSTRUCTION: Do NOT build a static dummy landing page. Build a fully functional, state-driven web application with real interactive filtering, live scoring algorithms, working modals, and dynamic data mutation across components. 1. TECH STACK & CLINICAL DESIGN SYSTEM: - Framework: Next.js 14 (App Router), TypeScript, Tailwind CSS, Lucide React, Framer Motion. - Fonts: 'Plus Jakarta Sans' (Headings), 'Inter' (Body), 'JetBrains Mono' (Costs, ICD-11 codes, GPS distances, Match Scores). - Color Palette: * Primary Background: #F8FAFC (Clinical Alabaster) * Card Surface: #FFFFFF with #E2E8F0 borders * Brand Primary: #0F172A (Deep Medical Navy) * Clinical Accent: #0D9488 (Trust Teal) * Verified/Operational: #10B981 (Emerald) * Warning/Maintenance: #F59E0B (Amber) * Critical/Unavailable/Emergency: #E11D48 (Coral Red) 2. CREATE THE CORE RELATIONAL DATA ENGINE (`src/data/medicalDatabase.ts`): Implement the exact relational structure with rich, realistic Indian healthcare data (12+ major hospitals across Kolkata, Delhi-NCR, Mumbai, Chennai, and Bengaluru including Tata Medical Center, SSKM Hospital, Apollo Multispeciality, AIIMS, Fortis, Medanta, Narayana Health, Manipal, and local nursing homes to demonstrate equipment mismatches): - `Hospitals` Schema: `hospital_id`, `name`, `hospital_type` ('Government' | 'Trust' | 'Corporate' | 'Nursing_Home'), `tier_category` ('Primary' | 'Secondary' | 'Tertiary_SuperSpecialty'), `address`, `city`, `pincode`, `coordinates` ({lat, lng}), `distance_km`, `eta_minutes`, `abdm_hfr_id`, `nabh_accredited`, `pmjay_empanelled`, `swasthya_sathi_accepted`, `emergency_24x7`, `contact_phone`, `last_verified_at`. - `Hospital_Equipment` Inventory: `id`, `hospital_id`, `equipment_code` ('PET_CT' | 'LINAC_RADIATION' | 'BRACHYTHERAPY' | 'CATH_LAB' | 'ECMO' | 'MRI_3T' | 'DIALYSIS_CRRT' | 'BIOPSY_HISTOPATH' | 'NEURO_MICRO' | 'VENTILATOR_ICU'), `equipment_name`, `brand_model`, `operational_status` ('OPERATIONAL' | 'MAINTENANCE' | 'UNAVAILABLE'), `total_units`, `available_units`, `avg_wait_time_hours`. - `Condition_Requirements` (AI Knowledge Graph): Map at least 6 major critical conditions (`2C25` Lung Cancer Stage IIA, `2C77` Carcinoma Cervix Stage IIB, `BA41` Acute STEMI Heart Attack, `8B11` Acute Ischemic Stroke, `GB61` End-Stage Renal Failure, `NC72` Severe Polytrauma) with `mandatory_equipment`, `optional_equipment`, `required_specialists`, `urgency_level`, and `cghs_base_rate`. - `Treatment_Costs`: `package_id`, `hospital_id`, `condition_icd_code`, `procedure_name`, `min_estimated_cost`, `max_estimated_cost`, `pmjay_package_rate`, `cashless_insurers`. 3. IMPLEMENT THE 50-25-25 CAPABILITY MATCH SCORING ALGORITHM (`src/utils/scoringEngine.ts`): Write a real TypeScript function `calculateHospitalMatchScore(hospital, condition, userFilters)` computing a live score (0 to 100%): - 50% Weightage: Percentage of `mandatory_equipment` currently `'OPERATIONAL'` at that hospital. - 25% Weightage: Financial & Scheme compatibility (PM-JAY / Swasthya Sathi / Budget match). - 25% Weightage: Travel ETA score + NABH Accreditation bonus. Wrap everything in a global React Context (`MedRadarContext`) so any filter, AI report, or admin machine toggle updates all hospital rankings in real time.
```

STEP 2 PROMPT: Sticky Clinical Navbar, Multilingual Switcher & Hero Command Center

```
Now build the Top Emergency Bar, Sticky Glassmorphic Navbar, and Interactive Hero Command Center connected to `MedRadarContext`. 1. TOP CLINICAL STATUS BAR & NAVBAR: - Top Strip: Live pulsing emerald indicator "ABDM HFR & NABH Equipment Grid Active | Real-Time Golden Hour Routing", quick switch for "B2B Hospital Admin Mode", and a working Multilingual UI Switcher (English | हिंदी / Hinglish | বাংলা). - Navbar: "MedRadar AI" logo with radar pulse icon, navigation links (AI Report Triage, Equipped Hospital Finder, 3-Tier Cost Calculator, Smart Route Map, Verify Hospital Portal), and a Coral Red CTA button: "🚨 SOS Golden Hour Triage". 2. INTERACTIVE HERO SECTION: - Headline: "Right Hospital. Right Machine. Right Cost — Before You Leave Home." - Interactive Dual-Mode Search Command Box: * Tab 1: "Search by Disease / Symptoms" (Typo-tolerant search supporting "kanser", "lung tumor", "hart atack", "stroke", "dialysis"). * Tab 2: "Search by Medical Equipment" (Checkboxes for PET-CT, Linear Accelerator, 24x7 Cath Lab, 3T MRI, ECMO, Ventilator ICU). * City & GPS Selector (Kolkata, Delhi-NCR, Mumbai, Bengaluru + "Simulate Live GPS"). * Quick Filter Chips: "Ayushman Bharat (PM-JAY)", "Swasthya Sathi Accepted", "Govt & Trust Only", "24x7 Cath Lab". - Right Column Live Comparison Card: Show "Nearby Unverified Nursing Home (2.1 km — 33% Match, Missing LINAC & PET-CT)" in Red vs. "Verified Tertiary Center (6.4 km — 98% Match, All Machines Operational)" in Emerald Green.
```

STEP 3 PROMPT: Working AI Medical Report OCR, DPDP Privacy Shield & Triage Studio

```
Now build Section 2: The "AI Multimodal Report Analyzer & Clinical Triage Engine" (`src/components/AiTriageStudio.tsx`). 1. LEFT PANEL — MULTIMODAL INPUT & DPDP PRIVACY ENGINE: - Include a working DPDP Act 2023 Privacy Shield checkbox: "Auto-Mask Patient PII (Name, Phone, Aadhaar) before AI Analysis". Show a visual redaction step when analyzing reports. - File Upload Zone + 4 One-Click Clinical Presets for instant demo testing: 1. "📄 Sample 1: Lung Biopsy & CT Chest Report (Oncology)" 2. "📄 Sample 2: Suspected Carcinoma Cervix Stage IIB (LINAC + Brachytherapy Needed)" 3. "📈 Sample 3: Acute Anterior Wall STEMI ECG (Golden Hour Cath Lab)" 4. "🧪 Sample 4: Renal Profile — Creatinine 8.4 mg/dL (Dialysis/CRRT)" - Voice & Hinglish Symptom Input Box (Bhashini Simulation): Allow typing or clicking sample Hindi/Hinglish voice notes like "Mere papa ko lung mein tumor aaya hai, doctor ne PET scan aur radiation bola hai". 2. RIGHT PANEL — HYBRID RAG + STRUCTURED JSON TRIAGE OUTPUT: - Include a toggle between `[Clinical Family View]` and `[Raw AI JSON & ICD-11 View]` showing `detected_condition`, `icd_11_code`, `urgency`, `required_diagnostics`, `required_treatment_infra`, and `required_specialists`. - Render the Mandatory Equipment Checklist, Required Super-Specialists under one roof, Clinical Referral Warning Box, and a working button: "Apply Triage Filter to Nearby Hospitals ->" that auto-filters and scrolls to the hospital grid.
```

STEP 4 PROMPT: Equipped Hospital Discovery Grid & Side-by-Side Comparison Matrix

```
Now build Section 3: The "Infrastructure-First Hospital Discovery & Comparison Engine" (`src/components/HospitalDiscoveryGrid.tsx`). 1. ADVANCED FILTER BAR & SORTING: - Filter by Specialty, Ownership (Govt / Trust / Corporate), Must-Have Operational Equipment pills, and Insurance/Scheme (PM-JAY, Swasthya Sathi, CGHS, Star Health). - Sort by: "Highest Capability Match Score (50-25-25 Algorithm)", "Fastest Ambulance ETA", or "Lowest Estimated Cost". 2. CLINICAL DASHBOARD HOSPITAL CARDS: Each card must display: - Verified NABH Badge, ABDM HFR ID, Ownership Tier, and Live Ambulance ETA (`6.4 km • 18 mins`). - Dynamic 50-25-25 Match Score Ring with hover breakdown (`Equipment: 50/50 | Budget/Scheme: 25/25 | Travel & NABH: 21/25`). - Live Equipment Inventory Matrix showing Green (`OPERATIONAL`), Amber (`MAINTENANCE`), or Red (`UNAVAILABLE`) badges for each required machine + waiting time. - 3-Tier Price & Scheme Box in JetBrains Mono font. - Actions: `[+ Compare]` checkbox (up to 3 hospitals), `[Inspect Machines & Pre-Arrival Checklist]` modal trigger, and `[📲 Send Summary on WhatsApp]` preview trigger. 3. WORKING COMPARISON DRAWER: Allow selecting up to 3 hospitals to open a side-by-side Comparison Matrix Modal comparing cost, scheme coverage, machine-by-machine status, wait times, ICU beds, and ETA.
```

STEP 5 PROMPT: 3-Tier Cost Estimation Calculator (CGHS & PM-JAY Multiplier Algorithm)

```
Now build Section 4: The "3-Tier Algorithmic Cost Estimator & Financial Aid Matcher" (`src/components/CostTransparencyEngine.tsx`). 1. INTERACTIVE CONTROLS: - Select Clinical Procedure, Room Category (General 1.0x, Semi-Private 1.35x, Private AC 1.75x), City Tier (Metro 1.0x vs Tier-2 0.8x), Financial Scheme (Self-Pay, PM-JAY, Swasthya Sathi, CGHS, Private Insurance), and a "15% ICU Contingency Buffer" toggle. 2. LIVE 3-TIER MULTIPLIER CARDS: Calculate and display live ranges using the official CGHS / PM-JAY HBP 2.2 baseline: - Tier 1: Government Hospital (`CGHS Base × 0.0x to 0.35x` — Free under PM-JAY/Swasthya Sathi). - Tier 2: Charitable Trust & Mid-Sized NABH (`CGHS Base × 1.5x to 2.2x` — Marked "Best Value & Fast Care"). - Tier 3: Corporate Super-Specialty (`CGHS Base × 3.0x to 4.8x`). 3. ITEMIZED BILL BREAKDOWN & FINANCING: Render a stacked breakdown bar (Equipment/Procedure 45%, ICU/Room 25%, Drugs/Stents/Consumables 20%, Doctor Fees 10%) plus a No-Cost Medical EMI & PMNRF Relief Grant eligibility estimator.
```

STEP 6 PROMPT: Geospatial Smart Rerouter Map & Pre-Arrival Checklist Modal

```
Now build Section 5: The "Geospatial Smart Rerouter & Hospital Infrastructure Deep-Dive" (`src/components/GeospatialRerouteMap.tsx`). 1. INTERACTIVE ROUTE SIMULATION MAP CANVAS: Show Patient's Home GPS Pin at the center with 3 interactive hospital pins: - Pin A (2.1 km — Local Nursing Home): Red Warning Marker explaining why visiting here wastes 90+ mins and ₹8,000 due to missing LINAC/Cath Lab. - Pin B (6.8 km — Verified Tertiary Center): Emerald Green Marker with animated direct route line (19 mins ETA, 100% Equipment Operational). - Pin C (9.4 km — Charitable Trust Alternative): Teal Marker showing 100% Equipment Match at 48% lower cost. 2. INFRASTRUCTURE & PRE-ARRIVAL CHECKLIST MODAL: Clicking any hospital opens a 3-tab modal: - Tab 1: Live Equipment Audit Table (Brand/Model, Status, Wait Time, and a crowdsourced "Was this machine working today? 👍 Yes / 👎 Report Downtime" button). - Tab 2: Disease-Specific Pre-Arrival Family Checklist (Aadhaar, Scheme Card, Biopsy Paraffin Block/Slides, Previous Reports). - Tab 3: Priority Triage Booking Form that generates a Verified Referral Token (`MR-2026-XXXX`).
```

STEP 7 PROMPT: Live B2B Hospital Admin Portal & SOS Golden Hour Emergency Mode

```
Now build the B2B Hospital Admin Dashboard and the SOS Golden Hour Emergency Responder Mode: 1. B2B HOSPITAL ADMIN & PRO DASHBOARD (`src/components/HospitalAdminModal.tsx`): - Allow selecting any hospital in our database and toggling its machines (`PET_CT`, `LINAC_RADIATION`, `CATH_LAB`, `ECMO`, `MRI_3T`) between `🟢 OPERATIONAL`, `🟡 MAINTENANCE`, and `🔴 UNAVAILABLE`, plus editing wait times and ICU bed counts. - REAL-TIME REACTIVITY: Switching a mandatory machine to `MAINTENANCE` must immediately recalculate that hospital's 50-25-25 Match Score on the main grid (dropping it from ~96% to ~60%) and flag an Amber "Machine Under Maintenance — Reroute Recommended" alert on its card. - Include a "Regional Demand Analytics (50 km Radius Search Trends)" SaaS preview inside the admin portal. 2. 🚨 SOS GOLDEN HOUR EMERGENCY MODE: Clicking the red Navbar SOS button opens a high-contrast emergency responder overlay with 1-tap triggers for: (1) Heart Attack STEMI (<60m Cath Lab), (2) Acute Brain Stroke (<4.5h Thrombolysis), (3) Severe Polytrauma, and (4) ECMO/Ventilator Crisis — immediately listing the Top 3 nearest 24x7 equipped hospitals sorted by Ambulance ETA with 1-click Call & Route actions.
```

STEP 8 PROMPT: WhatsApp Hindi Voice Bot Simulator, Legal Compliance & Final Polish

```
Final Step: Add the Floating WhatsApp Bot Simulator, Regulatory Compliance Footer, and polish all interactions. 1. FLOATING WHATSAPP TRIAGE BOT SIMULATOR (Bottom-Right Widget): - Open an interactive WhatsApp Business Cloud API + Bhashini Voice chat simulator where users can click sample voice/photo messages (e.g., "🎤 Hindi Voice Note: Mere mareez ko cancer radiation chahiye", "📄 Send Biopsy Photo", "📍 Share Live Location") and receive instant automated replies with the top 3 equipped hospitals, PM-JAY status, and estimated cost range. 2. REGULATORY, DATA TRUST & DPDP ACT 2023 FOOTER: - Display the 4-Layer Data Architecture badges (ABDM HFR, CGHS/PM-JAY HBP 2.2 Baselines, AI Vision Equipment Audit, Crowdsourced Hospital Verification). - Include the mandatory Clinical Disclaimer & DPDP Act 2023 Privacy Notice stating that MedRadar AI provides infrastructure discovery and algorithmic cost range estimates, not medical prescriptions.
```